# 1.1 Interaction of Heredity & Environment

**Everything in Psychology is simultaneously biological**

## Where do our traits come from?

The Nature vs. Nuture Debate has been active since Ancient times. Plato thought that people are born with inherent traits. (Nature) while Aristotle argued that our traits come from our environment. (nuture)

The current scientific consensus is that traits arise from the interaction between nature and nuture.

Charles Darwin proposed the evolutionary process of natural selection: nature favors certain traits that increase likelihood of survival, and those traits are more likely to get passed.

Evolutionary Psychology focuses on similarities between humans because of our shared biology & evolutionary history

Behavior Genetics focuses on differences that arise from our various genes & environments

## How does evolution affect our Behavior?

Evoloutionary Psychologists use natural selection to understand the roots of our behaviour & mental processes.

## Natural Selection

- Organisms' varied Offspring compete for survival.
- Certain traits ⇒ P(survive & reproduce) ↑
- Survive ⇒ P(pass on genes) ↑
- In this way, population characteristics may change

## Belyaev's Experiment

In the 1950s, the Soviet Union started an experiment that would demonstrate how we evolutionarily domesticated dogs.

Soviet scientist Belyaev started with 100 female & 30 male foxes. He bred them & selected the tamest fraction of their offspring to breed. Over 57 generations, he & his successor, trut, changed the foxes into docile, domesticated creatures.

Similarity to Belyaev, nature has selected certain variations to be more advantageous. These variations (+ mutations) cause us to evolve.

Compared to other animals, the human brain is far less restricted by its genes. It has the very powerful ability to adapt. This adaptability contributes to our fitness.

Humans share 99.9% of our DNA

$\leq 5\%$ of genetic differences arise from population group differences
~95% of genetic variation exists within groups

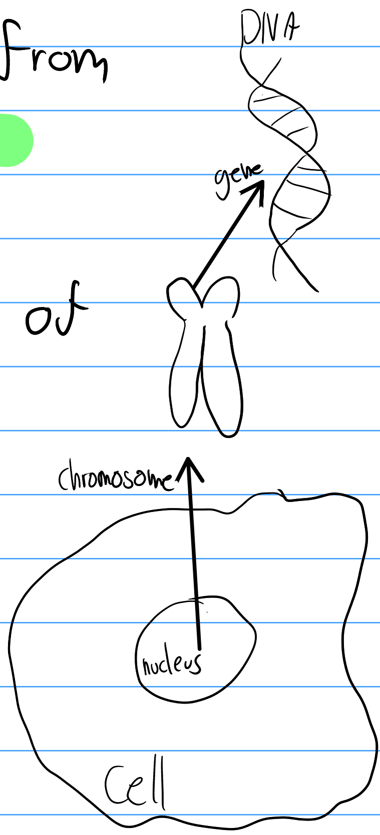
The genetic difference between two Indians is likely greater than the average genetic difference between Indians as a whole & (for ex.) Palestinians as a whole.

The natural dangers our psychology evolved to fight against no longer exist, sometimes leading our brains to turn those tools against us.

# What role do genes play in behavior

Behavior Geneticists explore the genetic and environmental roots of human differences

Our shared brain architecture leads us to share certain behaviors
But we are incredibly diverse.
Our diversity is shaped by our heridity & environment.




The instructions for your body come from (usually) 46 chromosomes - 23 from each parent

Each chromosome is a coiled chain of deoxyribonucleic acid - DNA

Genes are small segments of large DNA molecules.

You have ~20K protein-coding genes which are either active (expressed) or inactive. Environmental events can "turn on" genes, allowing them to code for proteins.

Researchers studying the human genome discovered the common sequence within human DNA. This shared gen. profile is what makes us humans.

Humans & Chimpanzees share 96% DNA & 99.4 % functional DNA.

That 0.6% contains the massive differences that seperate us from chimpanzees.

There is no single gene that predicts your smarts, sexual orientation or personality.

Our differing traits are poly genetic, influenced by many genes of small effect.

How can we learn more about nature vs nuture?

Family studies search for traits that tend to be shared by family members

Dizygotic (fraternal) twins develop from two seperate fertilized eggs.

They share no more DNA than regular siblings.

Monozygotic (identical) twins develop from a single fertilized egg that splits. They are gen.ly identical & always same-sex.

Not only do they share the same genes but also the same uterus, conception, & (usually) bday, culture

but the twins don't always share the same # of copies of each gene. They also sometimes differ in their brains' tiny wiring structures.

2/3 of mz twins share placenta

the diff. placentas of the other 1/3 contribute to diffs. between them

your mz twin has autism $\Rightarrow$ P(you have autism) = $\frac{3}{4}$

your dz twin has autism $\Rightarrow$ P(you have autism) = $\frac{1}{3}$

Mz twins are much more behaviourally alike.

MZ twins look alike, but that doesn't account for their behavioral similarities.

twin studies have demonstrated some similarities between mz twins raised in different environments, but its hard to account for various biases here.

Adoption makes 2 groups { Genetic Relatives
Environmental Relatives

The normal range of environments shared by a family's children has little discernable impact on their personcalities.

## Personal Connection

This is not a scientifically sound conclusion as it is built upon only 8 years of my experience + my parents anecdotal evidence, but I have noticed stark differences between what I know about myself as a young child & my now 8-year old sister, in terms of behavior.

## Adoption

Child abuse, neglect & divorce are far less common in adoptive families as adoptive parents are carefully screened.

Adoptive parents are gentler, give more guidance, & experience less depression

most adopted children thrive
many ad. children score higher than bio. parents & raised-apart bio. siblings.

# Nature and Nurture... ?

Some traits develop ~identically $\forall$ environments
Some adaptations require specific environments

Both genes & experience interact.

Epigenetics studies the molecular mechanisms by which env.s can trigger or block genetic expression.

Genes are self-regulating. Rather than acting as simple blueprints, genes react to the complex context.

Our experiences create epigenetic marks, often organic methyl molecules attached to part of a DNA strand.
If a mark instructs the cell to ignore any gene in a DNA segment, those genes will be 'turned off', stopping the proteins those genes code for from being produced.

"Things written in pen you can't change. That's DNA. But things written in pencil you can. That's epigenetics."

Genes

Prenatal ← Drugs, toxins, nutrition, stress

Postnatal ← Neglect, abuse, variation in care

Juvenile ← Social Contact, env.al complexity

Adult ← Cogn. challenges, exersize, nutrition

Gene expression affected by epigenetic molecules

Various environmental factors like diet, drugs, stress etc. can affect epigenetic markers.

Because Epigenetics, effects of childhood trauma & poverty may last a lifetime.

Some evo. biologists theorize inheritance also occurs through env. influences

Evolution Shapes Culture

Culture influences Evolution

## MCQs

1. C 2. D 3. C 4. A 5. B 6. D
7. C 8. B 9. D 10. A 11. D 12. C